

Exp No.8: Steering Angle Variation in MIMO Beamforming Using MATLAB

AIM: To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

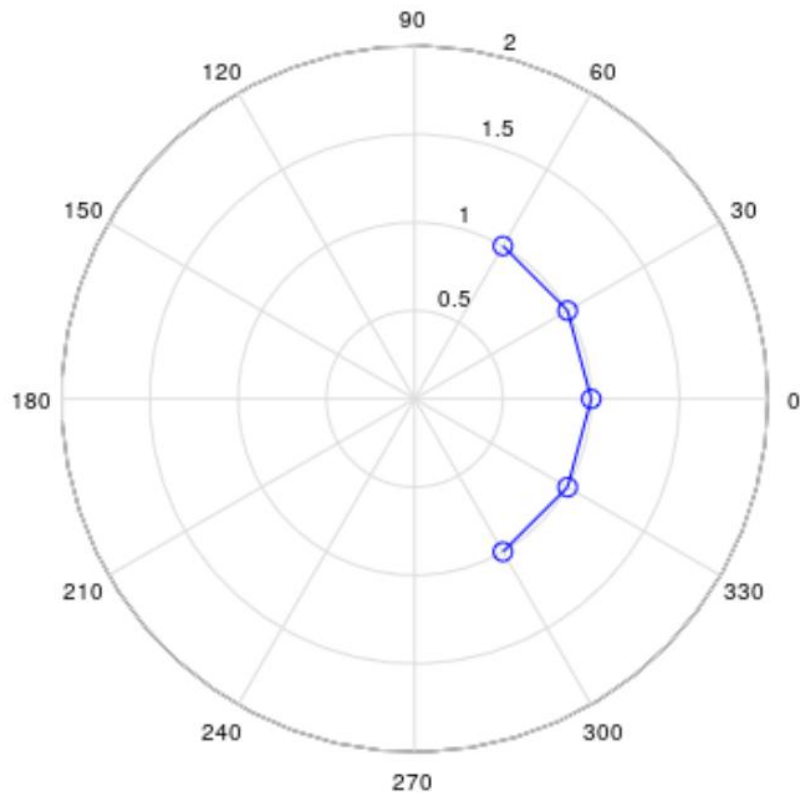
Objective 1: To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % Angles in degrees
6  theta = [-60 -30 0 30 60];
7
8  % Convert to radians
9  theta = theta*pi/180;
10
11 % Radius
12 r = ones(size(theta));
13
14 figure;
15 polar(theta,r,'ob');
16 hold on;
17 polar(theta,r,'-b');
18
19 title('192412380 Steering Angle Variation in MIMO Beamforming');
20
21 set(gca,'RTick',[0.5 1 1.5 2]);
```

OUTPUT :

192412380 Steering Angle Variation in MIMO Beamforming

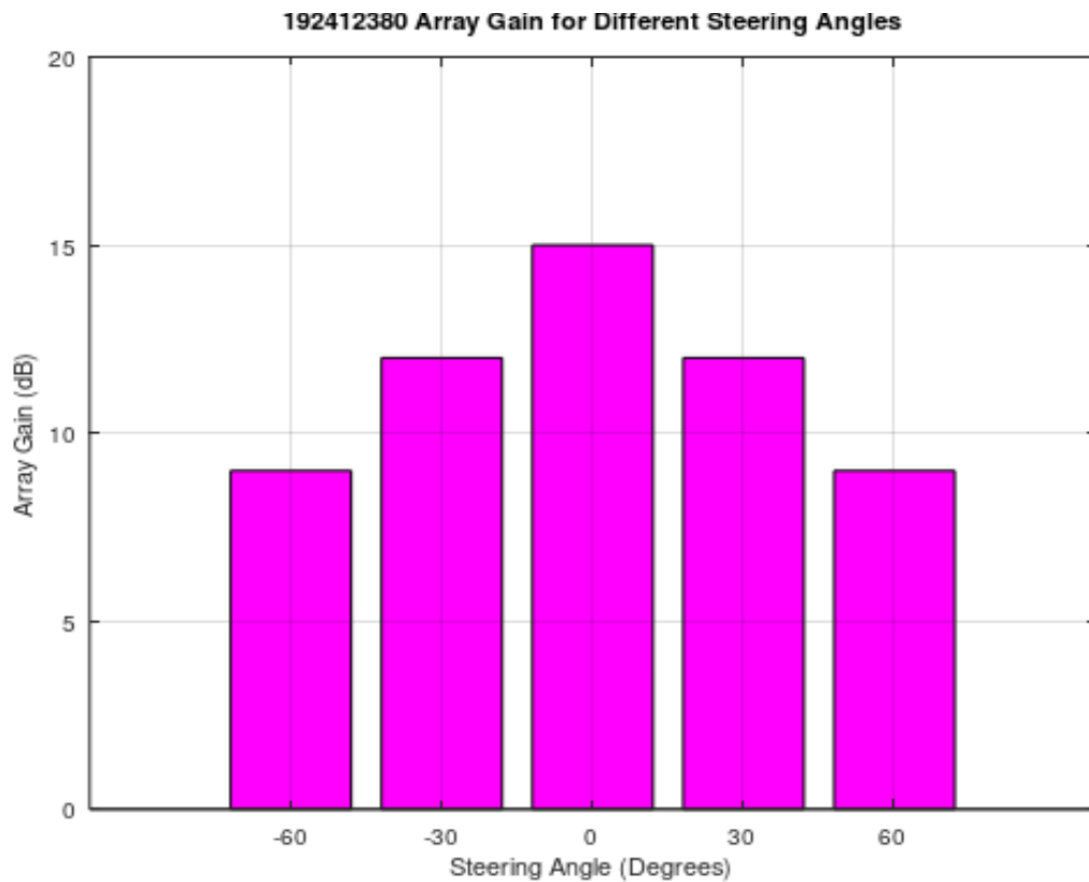


Objective 2: To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.

Program

```
1  clc;
2  clear;
3  close all;
4
5  % Steering Angles (Degrees)
6  angle = [-60 -30 0 30 60];
7
8  % Array Gain (dB)
9  gain = [9 12 15 12 9];
10
11 figure;
12
13 % Bar Graph (Magenta Color)
14 bar(angle, gain, 'm');
15
16 grid on;
17
18 xlabel('Steering Angle (Degrees)');
19 ylabel('Array Gain (dB)');
20 title('192412380 Array Gain for Different Steering Angles');
21
22 drawnow;
23 pause(5);
```

OUTPUT :

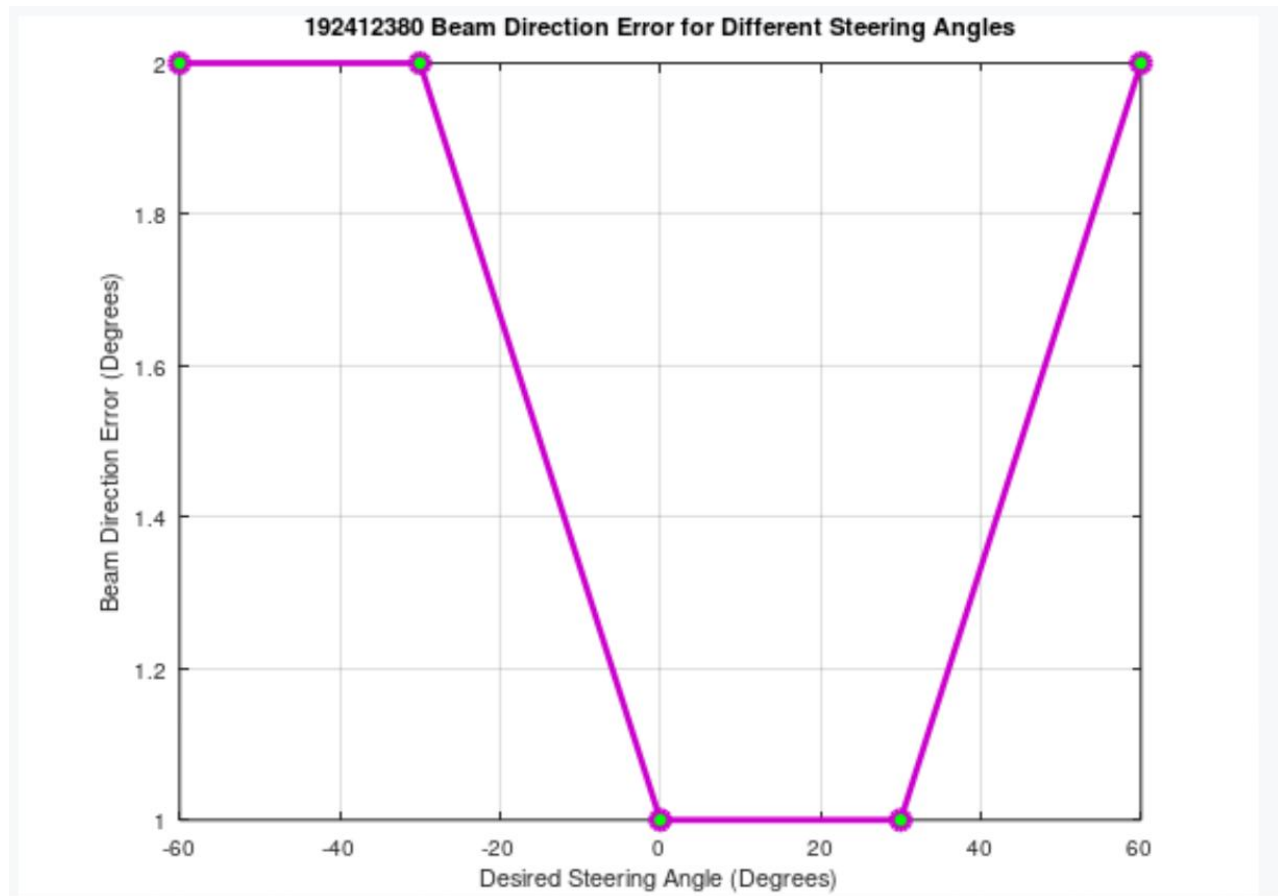


Objective 3: To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

program

```
1  clc;
2  clear;
3  close all;
4
5  % Desired Steering Angles (Degrees)
6  desired = [-60 -30 0 30 60];
7
8  % Actual Beam Directions (Degrees)
9  actual = [-58 -32 1 29 62];
10
11 % Beam Direction Error
12 error = abs(desired - actual);
13
14 % Display Results
15 fprintf('Desired\tActual\tError\n');
16 fprintf('-----\n');
17
18 for i = 1:length(desired)
19     fprintf('%d\t%d\t%d\n', desired(i), actual(i), error(i));
20 end
21
22 figure;
23
24 % Plot Beam Direction Error
25 plot(desired, error, ...
26     '-o', ...
27     'Color', [0.8 0 0.8], ...      % Purple line
28     'MarkerFaceColor', [0 1 0], ... % Green markers
29     'LineWidth', 2);
30
31 grid on;
32
33 xlabel('Desired Steering Angle (Degrees)');
34 ylabel('Beam Direction Error (Degrees)');
```

OUTPUT :



Result:

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.

Practical Questions

A wireless communication system employing MIMO beamforming with steering angle variation enables directional transmission by adjusting the phase of antenna elements to steer the beam toward desired user locations. This technique improves signal strength, coverage, and spatial accuracy in modern wireless systems such as 5G NR. Using MATLAB, simulate a MIMO beamforming system and analyze its performance.

1. To analyze the effect of steering angle variation on the direction of the main beam in a MIMO beamforming system using MATLAB.
2. To compare the array gain achieved at different steering angles and evaluate how beamforming improves signal strength.

3. To evaluate the beam direction error between the desired steering angle and the actual beam direction and analyze the accuracy of beam steering in a MIMO system.